



Outline

Optimization

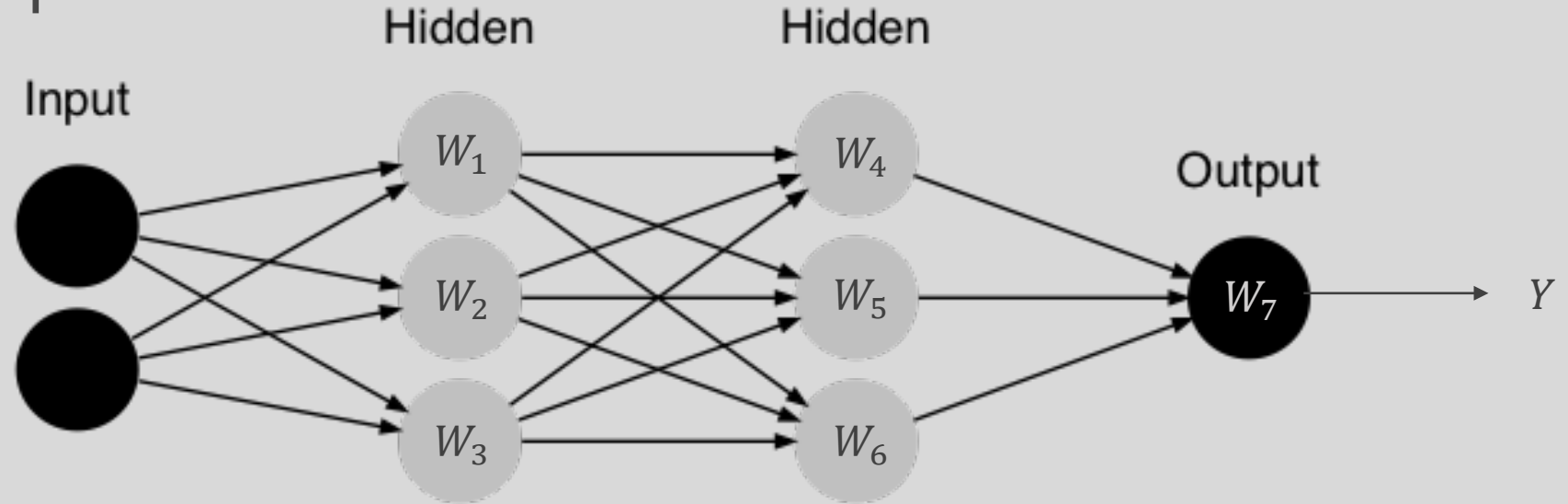
- Challenges in Optimization
- Momentum
- Adaptive Learning Rate
- Parameter Initialization
- Batch Normalization

Regularization of NN

- Norm Penalties
- Early Stopping
- Data Augmentation
- Sparse Representation
- Dropout



Sparse Representation

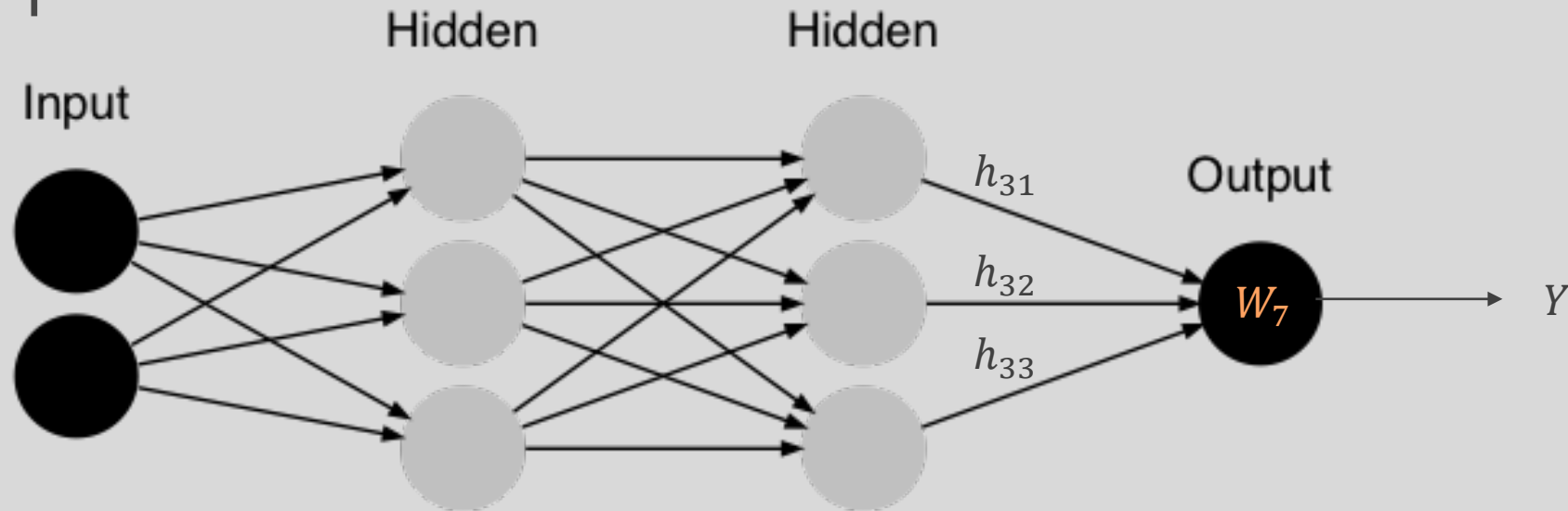


$$L(\theta; X, y)$$

$$[4.34] = \underbrace{[3.2 \quad 2.0 \quad 1.8]}_{W_7} \begin{bmatrix} 2 \\ -2.2 \\ 1.3 \end{bmatrix}$$



Sparse Representation

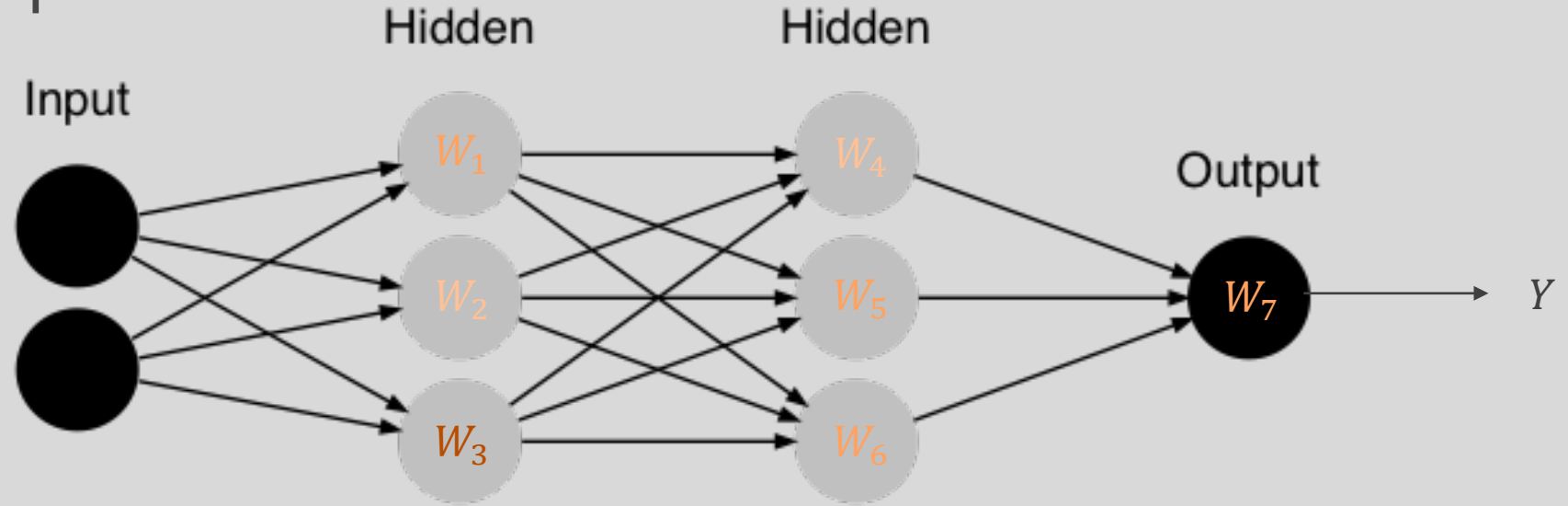


$$L(\theta; X, y)$$

$$[4.34] = [3.2 \quad 2 \quad 1] \left[\begin{array}{c} 2 \\ -2.2 \\ 1.3 \end{array} \right] \left. \vphantom{\begin{array}{c} 2 \\ -2.2 \\ 1.3 \end{array}} \right\} h_{31}, h_{32}, h_{33}$$



Sparse Representation



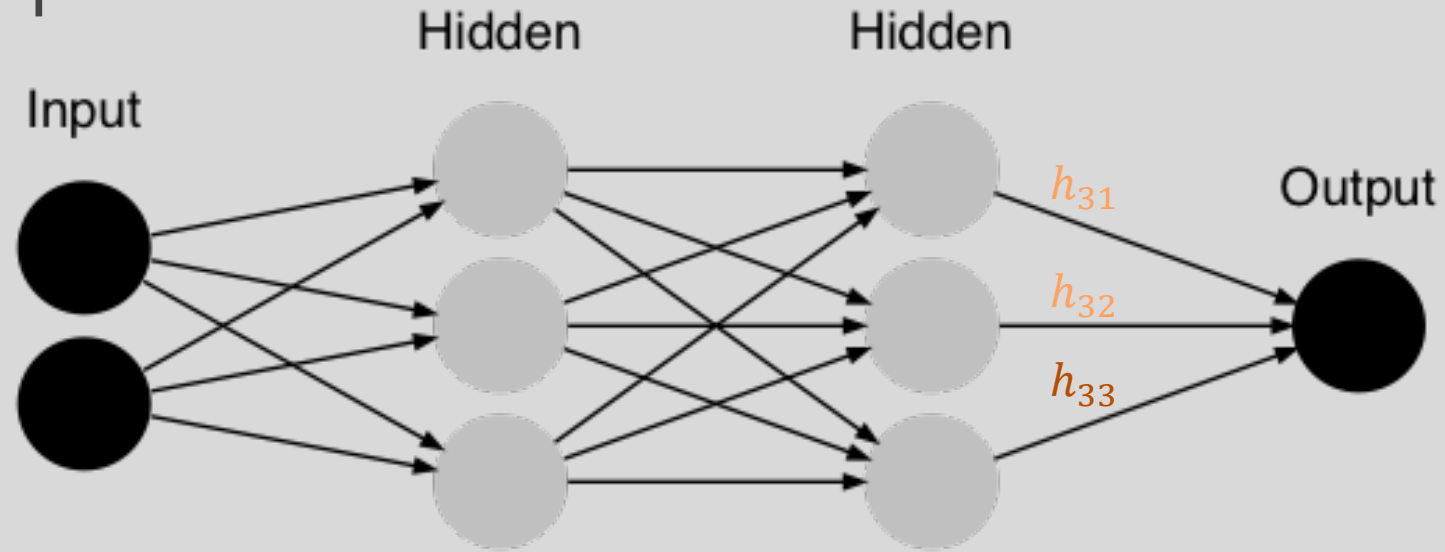
$$L_R(W; X, y) = L(\theta; X, y) + \alpha \Omega(W)$$

$$[0.69] = \underbrace{[0.5 \quad .2 \quad 0.1]}_{W_7} \begin{bmatrix} 2 \\ -2.2 \\ 1.3 \end{bmatrix}$$

Weights in output layer



Sparse Representation



$$L_R(W; X, y) = L(\theta; X, y) + \alpha \Omega(h)$$

$$[1.3] = [3.2 \quad 2 \quad 1] \begin{bmatrix} 0 \\ -0.2 \\ .9 \end{bmatrix} \quad \left. \vphantom{\begin{bmatrix} 0 \\ -0.2 \\ .9 \end{bmatrix}} \right\} h_{31}, h_{32}, h_{33}$$

Output of hidden layer



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Noise Robustness

Random **perturbation of network weights**

- Gaussian noise: Equivalent to minimizing loss with regularization term
- Encourages smooth function: small perturbation in weights leads to small changes in output

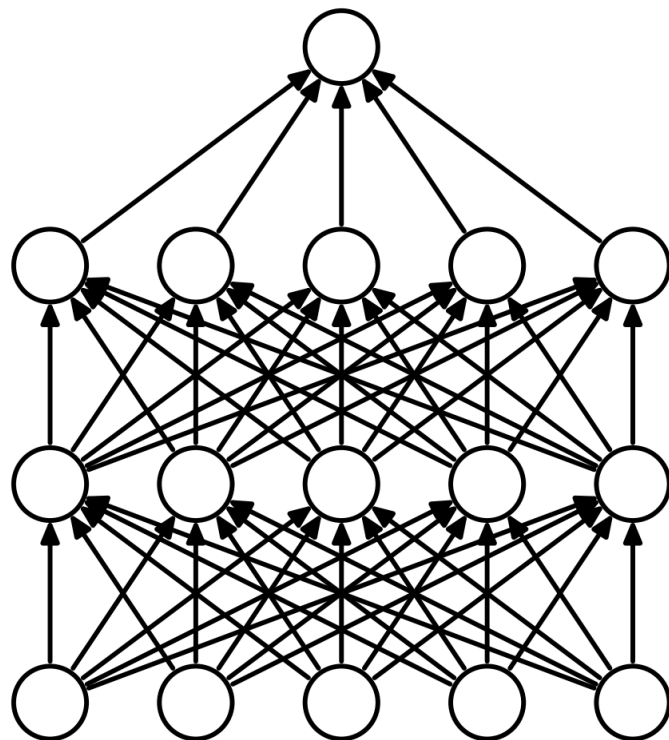
Injecting **noise in output labels**

- Better convergence: prevents pursuit of hard probabilities

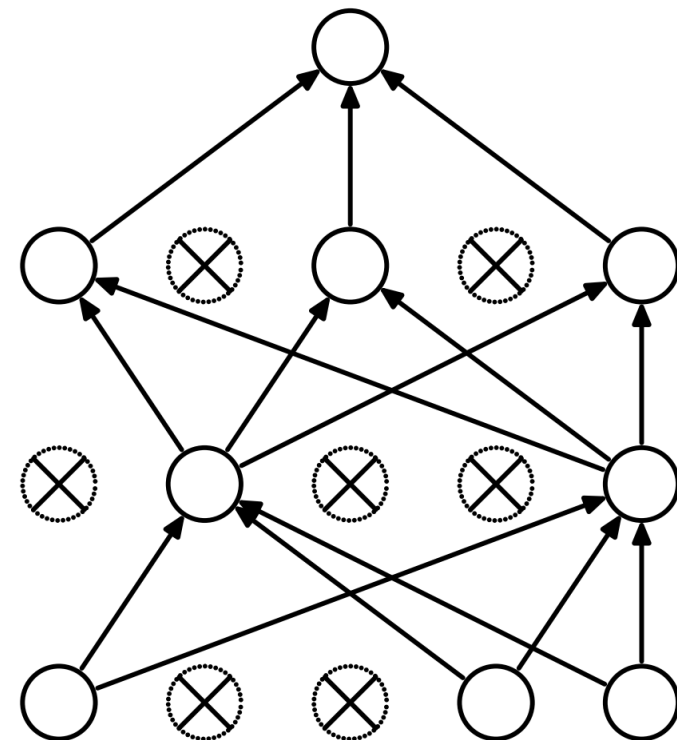


Dropout

- Randomly set some neurons and their connections to zero (i.e. “dropped”)
- Prevent overfitting by reducing co-adaptation of neurons
- Like training many random sub-networks



(a) Standard Neural Net

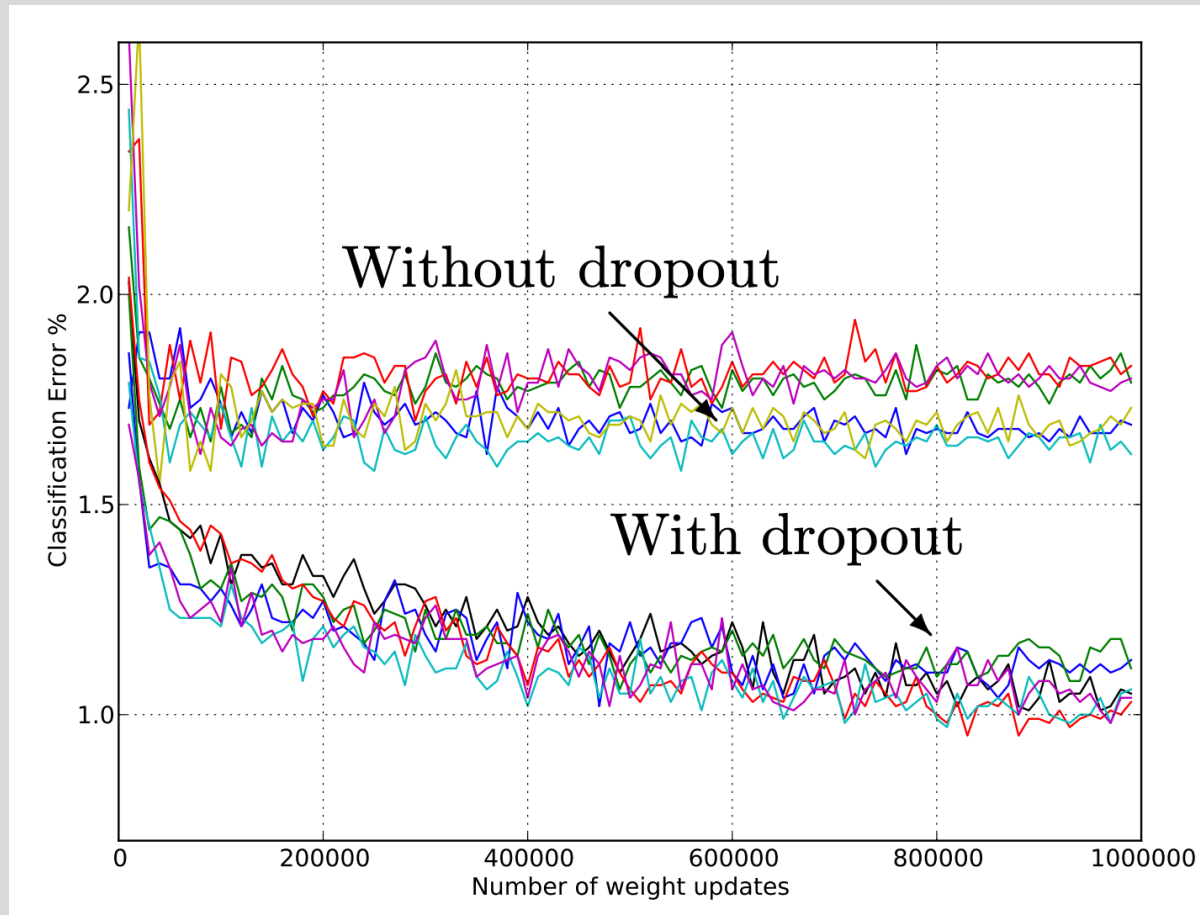


(b) After applying dropout.



Dropout

- Widely used and highly effective
- Proposed as an alternative to ensembling, which is too expensive for neural nets



Test error for different architectures with and without dropout. The networks have 2 to 4 hidden layers each with 1024 to 2048 units.

Image Source:

<http://jmlr.org/papers/volume15/srivastava14a/srivastava14a.pdf>



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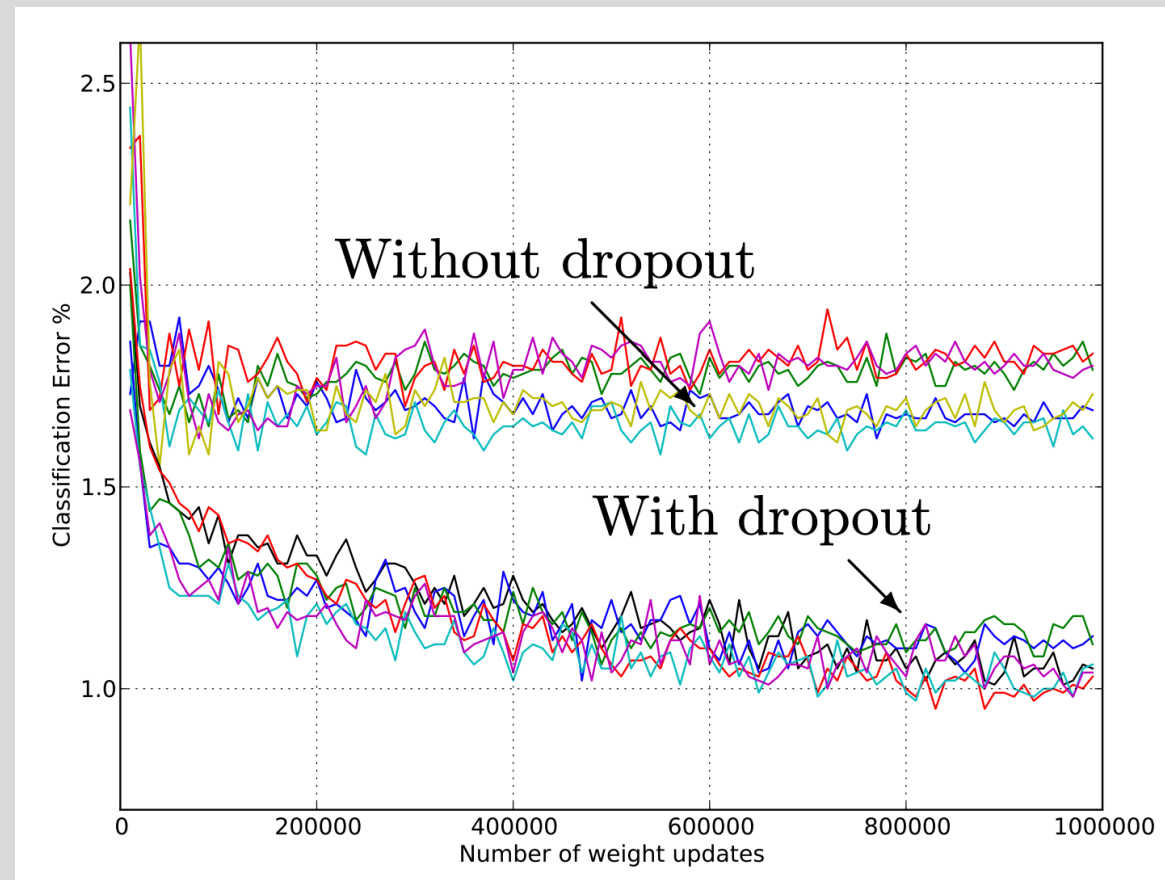
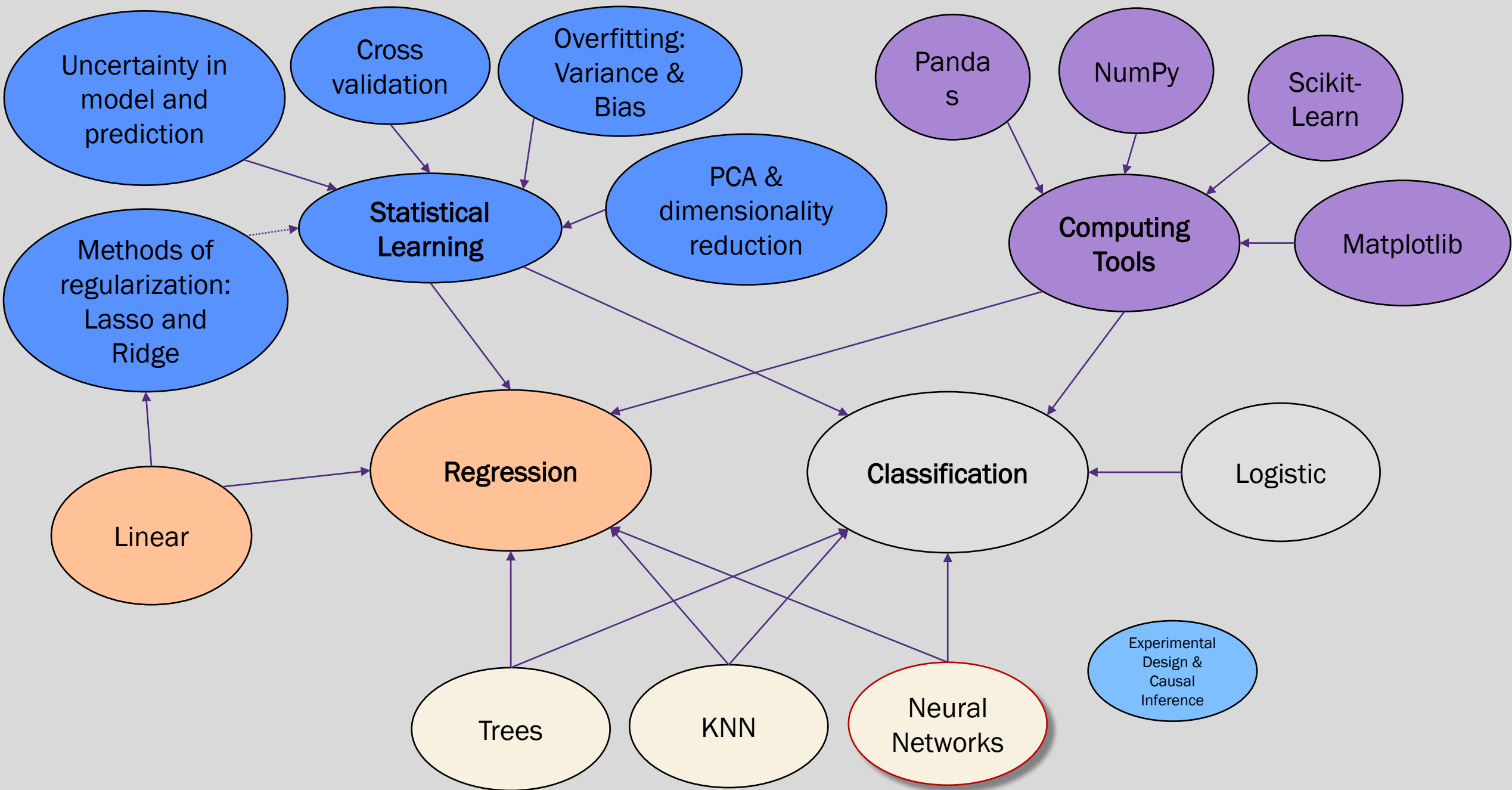


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Questions?



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